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VPC USING BASTION SERVER

1. VPC with Bastion Server

VPC & Subnet Design

VPC CIDR: 10.0.0.0/16

Public Subnet (Bastion Server): 10.0.1.0/24

Private Subnet (Database Server): 10.0.2.0/24

VPC > Your VPCs > Create VPC

Create VPC

Info

A VPC is an isolated portion of the AWS Cloud populated by AWS objects, such as Amazon EC2 instances.

VPC settings

Resources to create

Info

Create only the VPC resource or the VPC and other networking resources.

☒ VPC only

☐ VPC and more

Name tag - optional

Creates a tag with a key of 'Name' and a value that you specify.

MyVPC

IPv4 CIDR block

Info

☒ IPv4 CIDR manual input

☐ IPAM-allocated IPv4 CIDR block

IPv4 CIDR

10.0.0.0/16

CIDR block size must be between /16 and /28.

VPC > Subnets > Create subnet

Create subnet

Info

VPC ID

Create subnets in this VPC.

vpc-0012621a7a6f873f9 (MyVPC)

Associated VPC CIDRs

IPv4 CIDRs

10.0.0.0/16

Subnet settings

Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 1

Subnet name

Create a tag with a key of 'Name' and a value that you specify.

PublicSubnet(bastion)

The name can be up to 256 characters long.

VPC > Subnets > Create subnet

Subnet 1 of 1

Subnet name
Create a tag with a key of 'Name' and a value that you specify.
PrivateSubnet(database)
The name can be up to 256 characters long.

Availability Zone [Info](#)
Choose the zone in which your subnet will reside, or let Amazon choose one for you.
United States (N. Virginia) / use1-az2 (us-east-1a)

IPv4 VPC CIDR block [Info](#)
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.
10.0.0/16

IPv4 subnet CIDR block
10.0.2.0/24 256 IPs

Tags - optional

2. Security Group Configuration (VERY IMPORTANT)

Bastion Server Security Group (Public Subnet)

Allows SSH (22) from: My IP (Database Administrator)

EC2 > Security Groups > Create security group

Security group name [Info](#)
Bastion-SG
Name cannot be edited after creation.

Description [Info](#)
Bastion server access

VPC [Info](#)
vpc-0012621a7a6f873f9 (MyVPC)

Inbound rules [Info](#)

Type	Protocol	Port range	Source	Description - optional
SSH	TCP	22	My IP	183.82.97.138/32

Add rule

Database Server Security Group (Private Subnet)

Allows MySQL / Aurora (3306) from: custom 10.0.1.0/24 (Public subnet / Bastion subnet)

Allows SSH (22) from: custom Private IP of Bastion Server

Does NOT allow public access

EC2 > Security Groups > Create security group

Security group name [Info](#)
DB-SG
Name cannot be edited after creation.

Description [Info](#)
Database server access

VPC [Info](#)
vpc-0012621a7a6f873f9 (MyVPC)

Inbound rules [Info](#)

Type	Protocol	Port range	Source	Description - optional
MYSQL/Aurora	TCP	3306	Cust... 10.0.1.0/24	
SSH	TCP	22	Cust... sg-0168452ba3c6...	

3. Security Outcome

Database server is NOT publicly accessible

Only Bastion server can connect to DB server using SSH

External users cannot access DB directly

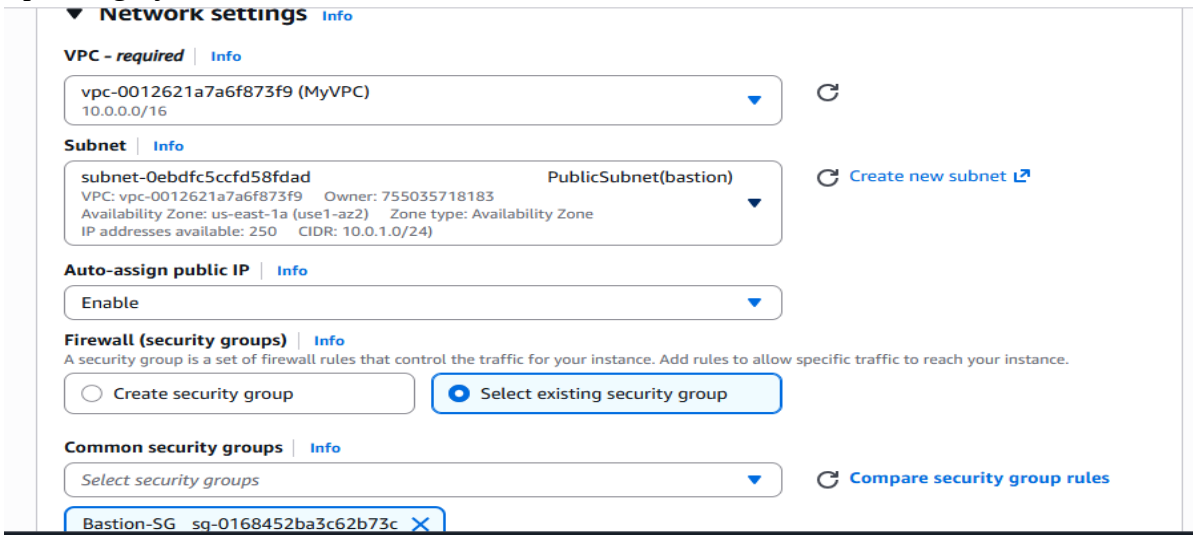
4. EC2 Instance Creation

Bastion Server

Subnet: Public Subnet (10.0.1.0/24)

Key pair: bastion.pem

Operating System: Ubuntu



The screenshot shows the 'Network settings' section of the AWS Management Console during EC2 instance creation. It includes fields for VPC (vpc-0012621a7a6f873f9), Subnet (subnet-0ebdfc5ccfd58fdad), and Auto-assign public IP (Enabled). The Firewall section shows 'Create security group' selected. The Common security groups section shows 'Bastion-SG' selected.

Network settings [Info](#)

VPC - required [Info](#)

vpc-0012621a7a6f873f9 (MyVPC)
10.0.0.0/16

Subnet [Info](#)

subnet-0ebdfc5ccfd58fdad PublicSubnet(bastion)
VPC: vpc-0012621a7a6f873f9 Owner: 755035718183
Availability Zone: us-east-1a (use1-az2) Zone type: Availability Zone
IP addresses available: 250 CIDR: 10.0.1.0/24

Auto-assign public IP [Info](#)

Enable

Firewall (security groups) [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☐ Create security group ☒ Select existing security group

Common security groups [Info](#)

Select security groups

Bastion-SG sq-0168452ba3c62b73c

Database Server

Subnet: Private Subnet (10.0.2.0/24)

Key pair: dbserver.pem

Operating System: Ubuntu

Name and tags [Info](#)

Name

Database-Server

[Add additional tags](#)

▼ Application and OS Images (Amazon Machine Image) [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Recents

Quick Start

Amazon Linux 	macOS 	Ubuntu 	Windows 	Red Hat 	SUSE Linux 	Del 	 Browse more AMIs Including AMIs from AWS, Marketplace and
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▼ Network settings [Info](#)

VPC - required [Info](#)

vpc-0012621a7a6f873f9 (MyVPC)
10.0.0.0/16



Subnet [Info](#)

subnet-0855e2cc02d7a9b4d PrivateSubnet(database)
VPC: vpc-0012621a7a6f873f9 Owner: 755035718183
Availability Zone: us-east-1a (use1-az2) Zone type: Availability Zone
IP addresses available: 250 CIDR: 10.0.2.0/24

[Create new subnet](#)

Auto-assign public IP [Info](#)

Disable

Firewall (security groups) [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☐ Create security group

☒ Select existing security group

Common security groups [Info](#)

Select security groups

[Compare security group rules](#)

DB-SG sq-0ddb6e547708851fe 

5. Copy DB Server Key to Bastion Server

Database server is in private subnet

Direct SSH from local system is NOT possible

DB server key must be copied to Bastion server

Command (PowerShell):

```
scp -i bastion.pem dbserver.pem ubuntu@<BASTION_PUBLIC_IP>:/home/ubuntu/
```

File is copied securely using SCP

dbserver.pem will now be available inside Bastion server

```
fs@DESKTOP-GGREQK6 MINGW64 ~/Desktop/aws (master)
$ scp -i bastion.pem dbserver.pem ubuntu@107.23.32.234:/home/ubuntu
dbserver.pem
```

6. Connect to Bastion Server

SSH Command:

ssh -i bastion.pem ubuntu@<BASTION_PUBLIC_IP>

Verify key files:

ls

Output:

bastion.pem

dbserver.pem

```
Connection to 107.23.32.234 closed.

fs@DESKTOP-GGREQK6 MINGW64 ~/Desktop/aws (master)
$ scp -i bastion.pem dbserver.pem ubuntu@107.23.32.234:/home/ubuntu
dbserver.pem

fs@DESKTOP-GGREQK6 MINGW64 ~/Desktop/aws (master)
$ ssh -i "bastion.pem" ubuntu@107.23.32.234
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-1018-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Tue Mar  3 06:26:53 UTC 2026

System load:  0.0           Temperature:   -273.1 C
Usage of /:   26.4% of 6.71GB Processes:    113
Memory usage: 24%          Users logged in: 0
Swap usage:   0%           IPv4 address for ens5: 10.0.1.175

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

Last login: Tue Mar  3 06:10:55 2026 from 183.82.97.138
ubuntu@ip-10-0-1-175:~$ ls
dbserver.pem
ubuntu@ip-10-0-1-175:~$
```

7. Connect from Bastion to Database Server

SSH from Bastion:

```
ssh -i dbserver.pem ubuntu@10.0.2.x
```

Connection to DB server is successful

Database administrator can now:

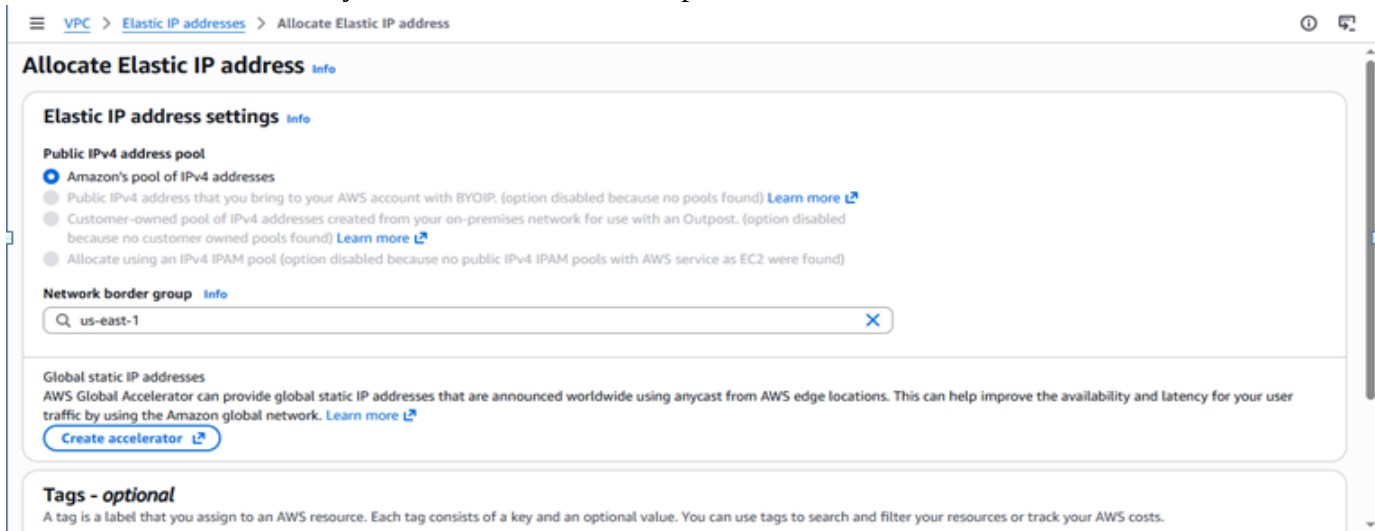
Update database configurations

Install required packages-----problem occurs

Restart MySQL service

VPC USING NAT GATEWAY

8. In the above steps it clear there is no internet access, if we want to install or update anything in the instances present in private subnet.
9. Network Address Translator (NAT Gateway) will solve this problem. Goto NAT Gateway in the left side menu bar of VPC and click on create NAT gateway.
10. Give the Nat gateway name, select your VPC, scroll down and click create NAT gateway.
11. So Now create a route for accessing NATGateway. Goto Route table click on create route table and give the route table name as bastion route , select our customized vpc and click on create route.
12. Select the route, click on actions, goto edit subnet associations and add private subnet to this bastion route.
13. Again goto actions and click on edit routes, attach NAT Gateway with 0.0.0.0/0, select your NAT gateway name in the drop down menu and click on save changes.
14. Now check in the dbserver you will internet in ec2 of private subnet.



Create NAT gateway [Info](#)

A highly available, managed Network Address Translation (NAT) service that instances in private subnets can use to connect to services in other VPCs, on-premises networks, or the internet.

NAT gateway settings

Name - optional

Create a tag with a key of 'Name' and a value that you specify.

The name can be up to 256 characters long.

Availability mode [Info](#)

Choose whether to deploy across all zones in the region or restrict to a single availability zone.

☒ Regional - new

Scales automatically across all regional AZs, simplifying management for multi AZ deployments.

☐ Zonal

Provides granular control within a specific availability zone, adhering to subnet level settings.

VPC

Select a VPC in which to create the regional NAT gateway.

Connectivity type

Select a connectivity type for the NAT gateway.

☒ Public☐ Private

Subnets

Route tables

Internet gateways

Egress-only internet gateways

Carrier gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Peering connections

Route servers

▼ Security

Network ACLs

Security groups

▼ PrivateLink and Lattice

Getting started

Endpoints

Route tables (1/4) [Info](#)

Last updated less than a minute ago

[Actions](#)[Create route table](#)

☒	Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC
☒	Private-RT	rtb-0c3a4f7cfea0e6d32	subnet-0e08c6cfb46c5d5...	-	No	vpc
☐	Public-RT	rtb-00ccc1cb63727da77	subnet-006eced88f82f8d...	-	No	vpc
☐	-	rtb-0cfe50e6f405abaf6	-	-	Yes	vpc
☐	-	rtb-02147bffd9a85dfb3	-	-	Yes	vpc

rtb-0c3a4f7cfea0e6d32 / Private-RT

[Details](#)[Routes](#)[Subnet associations](#)[Edge associations](#)[Route propagation](#)[Tags](#)

Details

Route table ID

rtb-0c3a4f7cfea0e6d32

VPC

Main

No

Owner ID

o-00000000000000000000000000000000

Explicit subnet associations

subnet-0e08c6cfb46c5d584 / Private-Subnet

Edge associations

-

Edit routes

Destination

10.0.0.0/16

Target

local

Status

Active

Propagated

No

Route Origin

CreateRouteTable

NAT Gateway

-

No

CreateRoute

[Remove](#)[Add route](#)[Cancel](#)[Preview](#)[Save changes](#)

```

Get:45 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [973 kB]
Get:46 http://security.ubuntu.com/ubuntu noble-security/universe Translation-en [218 kB]
Get:47 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [74.3 kB]
Get:48 http://security.ubuntu.com/ubuntu noble-security/universe amd64 c-n-f Metadata [20.6 kB]
Get:49 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [2568 kB]
Get:50 http://security.ubuntu.com/ubuntu noble-security/restricted Translation-en [596 kB]
Get:51 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Components [212 B]
Get:52 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 c-n-f Metadata [536 B]
Get:53 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Packages [28.8 kB]
Get:54 http://security.ubuntu.com/ubuntu noble-security/multiverse Translation-en [6732 B]
Get:55 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Components [208 B]
Get:56 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 c-n-f Metadata [396 B]
Fetched 40.4 MB in 8s (5136 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
99 packages can be upgraded. Run 'apt list --upgradable' to see them.
ubuntu@ip-10-0-2-113:~$

```

SCENARIO BASED QUESTIONS

1. You have multiple private subnets in different AZs. Should you use one NAT Gateway or multiple NAT Gateways? Explain your design choice.

Ans: Use one NAT Gateway per AZ for high availability and to avoid cross-AZ data charges. This ensures redundancy if an AZ fails.

2. Your private subnet instances can reach the internet, but the NAT Gateway shows no traffic. What could be the possible reasons?

Ans: Instances may have public IPs and bypass NAT, or route tables are misconfigured. Traffic might be going through another device or direct IGW.

3. A NAT Gateway was accidentally created in a private subnet instead of a public subnet. What problem will occur, and how would you fix it?

Ans: NAT Gateway won't have internet access, breaking outbound traffic from private instances. Fix by deleting and recreating it in a public subnet.

4. An application running in a private subnet requires outbound access only to a few trusted external services. How would you restrict traffic while still using a NAT Gateway?

Ans: Use security groups, network ACLs, or proxy servers to restrict outbound traffic to trusted IPs/domains. VPC endpoints can also limit access to AWS services.

5. Multiple teams share a single VPC and NAT Gateway. One team's traffic is affecting others. How would you redesign the NAT Gateway setup to isolate traffic?

Ans: Create separate NAT Gateways per team/subnet with dedicated route tables. This isolates traffic and improves control and billing.

6. You can successfully SSH into a private EC2 instance using a Bastion server, but yum update / apt update fails inside the private instance. Why?

Ans: The private instance lacks internet access due to missing NAT Gateway or incorrect route table settings.

7. Private EC2 instances can access the internet through a NAT Gateway, but administrators cannot SSH into them using the Bastion server.

Ans: Security groups or network ACLs likely block SSH from Bastion, or Bastion's network setup is incorrect.

8. A Bastion server and NAT Gateway are both created in a private subnet by mistake. What impact does this cause?

Ans: Both won't have internet access, breaking SSH connectivity and outbound internet for private instances.

9. SSH access to the Bastion server works, but SSH from Bastion to the private EC2 fails. Why?

Ans: The private EC2's security group or ACLs block SSH from Bastion, or wrong SSH credentials are used.

10. A student claims that a Bastion server alone is enough to provide internet access to private instances. Why is Bastion not a replacement for NAT Gateway?

Ans: Bastion only provides SSH access; it does not route or translate outbound internet traffic like a NAT Gateway.